



NumPy Tricks & Functions Lab

Sorting, Concatenation, Unique, Where & More

Based on: *3.Numpy-trick.pdf*

55 Hands-on Programming Questions (No Theory)

◀▶ Solve by coding only — each question tests a specific NumPy function.



BASIC: np.sort() - Sorting Arrays

? Question Create an array `arr = np.array([5, 2, 8, 1, 9, 3])` and sort it using `np.sort()`. Print both the original and sorted arrays to show the original remains unchanged.

? Question Sort a 2D array `arr_2d = np.array([[3, 1, 4], [6, 2, 5]])` row-wise (`axis=1`) and column-wise (`axis=0`).

? Question Demonstrate the difference between `np.sort(arr)` and `arr.sort()` using `arr = np.array([3, 1, 2])`.

? Question Sort the array `arr = np.array([5, 2, 8, 1, 9, 3])` using different algorithms: 'quicksort', 'mergesort', and 'heapsort'.

? Question Create a 2D array of shape (3,4) with random integers and sort it along `axis=0` and `axis=1` separately.



BASIC: np.append() - Adding Elements

? Question Given `arr = np.array([1, 2, 3])`, append the value 4 to it using `np.append()`. Show that the original array remains unchanged.

? Question Given `arr_2d = np.array([[1, 2, 3], [4, 5, 6]])`, append a new row `[7, 8, 9]` using `axis=0`.

? Question Append a new column `[[10], [11]]` to the same 2D array using `axis=1`.

? **Question** Demonstrate what happens when you use `np.append()` without specifying the axis parameter on a 2D array.

? **Question** Measure and compare the time complexity of appending elements in a loop using `np.append()` vs pre-allocating an array.

☰ 🔗 BASIC: `np.concatenate()` - Joining Multiple Arrays

? **Question** Concatenate three 1D arrays `arr1 = np.array([1, 2, 3])`, `arr2 = np.array([4, 5, 6])`, and `arr3 = np.array([7, 8, 9])` using `np.concatenate()`.

? **Question** Vertically concatenate (`axis=0`) two 2D arrays `arr1 = np.array([[1, 2, 3], [4, 5, 6]])` and `arr2 = np.array([[7, 8, 9], [10, 11, 12]])`.

? **Question** Horizontally concatenate (`axis=1`) the same two 2D arrays.

? **Question** Use `np.vstack()` and `np.hstack()` as convenience functions for the same operations.

? **Question** Try to concatenate arrays with mismatched shapes and observe the error message.

☰ ★ INTERMEDIATE: `np.unique()` - Finding Distinct Values

? Question Given `arr = np.array([1, 3, 2, 3, 1, 2, 4, 2, 1, 5])`, find all unique values using `np.unique()`.

? Question Find the unique values and their counts using `return_counts=True` for the same array.

? Question Use `return_inverse=True` to reconstruct the original array from unique values.

? Question Given a 2D array `arr_2d = np.array([[1, 2, 1], [2, 3, 2], [1, 2, 1]])`, find the unique rows using `axis=0`.

? Question Implement label encoding for `categories = np.array(['red', 'blue', 'red', 'green'])` using `np.unique()` with `return_inverse=True`.

? Question Use `np.unique()` to remove duplicate values from a large array and measure the time compared to using Python's `set()`.

☰ Q INTERMEDIATE: `np.where()` - Conditional Selection

? Question Given `arr = np.array([1, 2, 3, 4, 5, 6])`, use `np.where()` to find the indices where values are greater than 3.

? Question Replace values in `arr` where condition is True with the square of the value, otherwise keep the original value.

? Question Clean a temperature dataset `temps = np.array([22, 25, -999, 28, -999, 24])` by replacing -999 with `np.nan` using `np.where()`.

? Question Use `np.where()` with multiple conditions on `arr_2d = np.array([[1, 2], [3, 4]])` to label values as 'High' (`i>2`) or 'Low' (`i<2`).

? Question Create a function that uses `np.where()` to clip values in an array between a lower and upper bound.

☰ ↑ INTERMEDIATE: `np.argmax()` and `np.argmin()`

? Question Given `arr = np.array([10, 25, 5, 30, 15])`, find the index of the maximum and minimum values using `np.argmax()` and `np.argmin()`.

? Question For a 2D array `arr_2d = np.array([[8, 3, 5], [1, 9, 4], [2, 7, 6]])`, find the index of the maximum value in each row (`axis=1`).

? Question Find the index of the maximum value in each column (`axis=0`) for the same 2D array.

? Question Find the overall maximum value and its flattened index in the 2D array.

? Question Use `np.argmax()` to find the best performing model from an array of accuracy scores.

☰ 📈 INTERMEDIATE: `np.cumsum()` - Cumulative Sum

? Question Given `arr = np.array([1, 2, 3, 4, 5])`, compute the cumulative sum using `np.cumsum()`.

? Question For `arr_2d = np.array([[1, 2, 3], [4, 5, 6]])`, compute the cumulative sum along rows (`axis=1`) and along columns (`axis=0`).

? Question Compute the flattened cumulative sum of the same 2D array.

? Question Use `np.cumsum()` to compute the running total of daily sales data.

? Question Implement a function that uses `np.cumsum()` to compute the prefix sum of an array.

≡ % INTERMEDIATE: np.percentile() - Percentile Calculation

? Question Given `data = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10])`, compute the median (50th percentile) using `np.percentile()`.

? Question Compute the 25th and 75th percentiles of the same data.

? Question Calculate multiple percentiles [25, 50, 75] at once.

? Question For a 2D array `arr_2d = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])`, compute the median along `axis=0` and `axis=1`.

? Question Calculate the Interquartile Range (IQR) for the dataset.

⌵ ADVANCED: np.histogram() - Histogram Computation

? **Question** Given `data = np.array([1, 2, 2, 3, 3, 3, 4, 4, 5])`, compute the histogram with 5 bins using `np.histogram()`. Print both counts and bin edges.

? **Question** Specify custom bin edges `[1, 2, 3, 4, 5, 6]` for the same data.

? **Question** Use `np.histogram()` with range parameter `range=(1, 5)` for the same data.

? **Question** Generate 1000 random numbers from a normal distribution and compute the histogram with 20 bins.

? **Question** Create a function that uses `np.histogram()` to detect outliers in a dataset.

⌵ ADVANCED: Function Comparisons

? **Question** Compare `np.concatenate()`, `np.vstack()`, and `np.hstack()` by using all three to combine the same arrays.

? **Question** Compare `np.append()` vs `np.concatenate()` in terms of:

- (a) Syntax
- (b) Performance (use time module)
- (c) Use cases

? Question Compare `np.where()` vs boolean indexing for conditional selection in terms of:

- (a) Readability
- (b) Performance
- (c) Use cases



ADVANCED: Best Practices & Mixed Concepts

? Question Demonstrate why using `np.append()` in a loop is inefficient. Create a loop that appends 1000 elements and measure the time.

? Question Show how to avoid the `np.append()` loop issue by pre-allocating an array of known size.

? Question Create a function that uses `np.where()` to replace outliers in a dataset with the median value.

? Question Use `np.unique()` with `return_counts=True` to create a frequency distribution table from a categorical dataset.

? Question Implement a function that takes a 2D array and:

- (a) Sorts it along `axis=0`
- (b) Finds unique values
- (c) Computes cumulative sum
- (d) Returns percentiles

? Question Write a complete data preprocessing pipeline using NumPy functions that:

- (a) Handles missing values using `np.where()`
- (b) Removes duplicates using `np.unique()`
- (c) Normalizes the data
- (d) Sorts the data
- (e) Splits the data

? Question Create a function that takes an array and a threshold value, and uses `np.where()` to return:

- (a) Values above threshold
- (b) Values below threshold
- (c) Indices of values above threshold

? Question Implement a function using `np.cumsum()` to compute the moving average of a time series with a given window size.

✔ Ready to Code! ✔

“Master these functions — they are the building blocks of every data science pipeline.”

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